

Module-1

Lecture-1: Introduction

MECHANICS OF SOLIDS

- ① Rattan, Strength of Materials, 2e McGraw Hill Education India, 2011
- ② S. H. Crandal, N. C. Dhal, T. J. Lardner, An introduction to the Mechanics of Solids, McGraw Hill, 1999
- ③ I.H. Shames, J. H. Pitarresi, Introduction to Solid Mechanics, Prentice Hall of India, 2006
- ④ R. C. Hibbeler, Mechanics of Materials, Pearson Education, 2008
- ⑤ James M. Gere, Stephen Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, New Delhi, 2012
- ⑥ F. Beer, E. R. Johnston, J. T. DeWolf, Mechanics of Materials, Tata McGraw Hill, 2011
- ⑦ A. Pytel, F. L. Singer, Strength of Materials, Harper & Row Publishers, New York, 1998
- ⑧ R. K. Bansal, Mechanics of solids, Laxmi Publications, 2004
- ⑨ P. N. Singh, P. K. Jha, Elementary Mechanics of Solids, Wiley Eastern Limited, 2012

Strength of Materials

Rigid body

A body which doesn't deform under the action of external force

Strength

Ability to resist failure and behaviour under the action of external forces

Strength of Materials

Engineering stream which deals with the behavior of solid objects subject to stresses and strains

Stephen Timoshenko

- **Stephen Timoshenko** (1878 - 1972)
- A Russian born, American engineer and academician
- Father of modern engineering mechanics
- His seminal works in the areas of engineering mechanics, elasticity and strength of materials, many of which are still widely used today
- Timoshenko beam theory



Refer:

<https://engineering.stanford.edu/about/heroes/stephen-timoshenko>

<http://www.nasonline.org/publications/biographical-memoirs/memoir-pdfs/timoshenko-stephen-1.pdf>

Basic Terminologies

Elasticity

- Physical bodies deform on external force
- Ability of a body to regain its original size and shape when external force is removed is called Elasticity

Stress

- Resistance force offered by a body against deformation measured per unit area is called Stress (σ)
- Mathematically $\sigma = \frac{P}{A}$
- SI Unit of stress is Pascal (Pa) which is equal to $1N/m^2$

Strain

- Deformation of a body per unit length is called Strain (ϵ)
- Mathematically $\epsilon = \frac{\delta l}{l}$
- Dimensionless quantity

Unit Conversion

- $10^{-12} = \text{pico (p)}$
- $10^{-9} = \text{nano (n)}$
- $10^{-6} = \text{micro } (\mu)$
- $10^{-3} = \text{milli (m)}$
- $10^3 = \text{Kilo (k)}$
- $10^6 = \text{Mega (M)}$
- $10^9 = \text{Giga (G)}$
- $10^{12} = \text{Tera (T)}$

*Thank
you!*